

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTAIOE705B - BLOCK CHAIN TECHNOLOGY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Blockchain achieves trust in a trustless environment primarily through:
a) Cryptographic verification and distributed consensus
b) Central authority validation c) Legal enforcement mechanisms
d) High transaction throughput
- 2 Collision resistance in a hash function implies that it is computationally infeasible to:
a) Generate a hash b) Find two different inputs with the same hash
c) Encrypt plaintext d) Decrypt ciphertext
- 3 SHA-256 generates a hash value of length:
a) 128 bits b) 160 bits c) 256 bits d) 512 bits
- 4 A Bitcoin wallet is mainly used to:
a) Store private and public keys b) Mine bitcoins
c) Validate blocks d) Execute smart contracts
- 5 A consortium blockchain differs from a private blockchain because it:
a) Is permissionless b) Is fully public c) Allows anonymous mining
d) Is governed by multiple organizations
- 6 The Ethereum Virtual Machine (EVM) is responsible for:
a) Mining Ether b) Managing wallets
c) Executing smart contracts d) Creating blocks
- 7 Which algorithm is commonly used in Bitcoin for digital signatures?
a) RSA b) DSA c) Diffie-Hellman d) ECDSA
- 8 The security strength of SHA-256 primarily relies on its:
a) Speed of execution b) Block size
c) Output length and avalanche effect d) Encryption capability
- 9 A hard fork results in:
a) Temporary network slowdown b) Backward compatibility
c) Permanent chain split d) Reduced mining rewards

- 10 Bitcoin mining contributes to network security by:
- Increasing transaction speed
 - Making block alteration computationally expensive
 - Encrypting wallet addresses
 - Reducing network latency
- 11 Cross-border payments using blockchain are beneficial because they are:
- More expensive
 - Faster and cost-effective
 - Completely anonymous
 - Restricted to banks
- 12 In a Merkle tree, changing a single transaction affects:
- All hashes up to the root
 - Only the leaf node
 - Only the sibling node
 - Only the block header
- Q2. Solve the following.
- Differentiate between blockchain systems using any six points, including public, private, and consortium blockchains, with real-world use cases. 6
 - Define blockchain and explain its key characteristics in detail with suitable examples. 6
- Q3. Solve the following.
- List the properties of a cryptographic hash function. Explain the working principle of SHA-256 with a neat diagram. 6
 - Differentiate between blockchain systems using any six points, including public, private, and consortium blockchains, with real-world use cases. 6
- Q4. Solve Any Two of the following.
- Define Bitcoin and explain the purpose of the genesis block. Explain, with examples, hard fork and soft fork scenarios. 6
 - Explain hard forks and soft forks in detail, using examples to show how they work. 6
 - Define unspent transaction outputs (UTXO). Explain how UTXOs are used to validate Bitcoin transactions. 6
- Q5. Solve Any Two of the following.
- Describe the basic components of the Ethereum ecosystem. Compare Ethereum and Bitcoin with respect to programmability and functionality. 6
 - Describe the Ethereum Virtual Machine (EVM) and its execution model. Describe the basic components of the Ethereum ecosystem. 6
 - Write a simple Solidity smart contract for transferring Ether between two accounts. 6
- Q6. Solve Any Two of the following.
- Describe the role of tokenization in supply chain financing. Describe the role of blockchain in secure electronic medical records. 6
 - Identify the stakeholders involved in blockchain-based land registry systems, and how their roles, interests, and power dynamics influence system design and outcomes? 6
 - Describe the objectives of Project Ubin. Explain how blockchain enhances transparency in supply chain management. 6

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